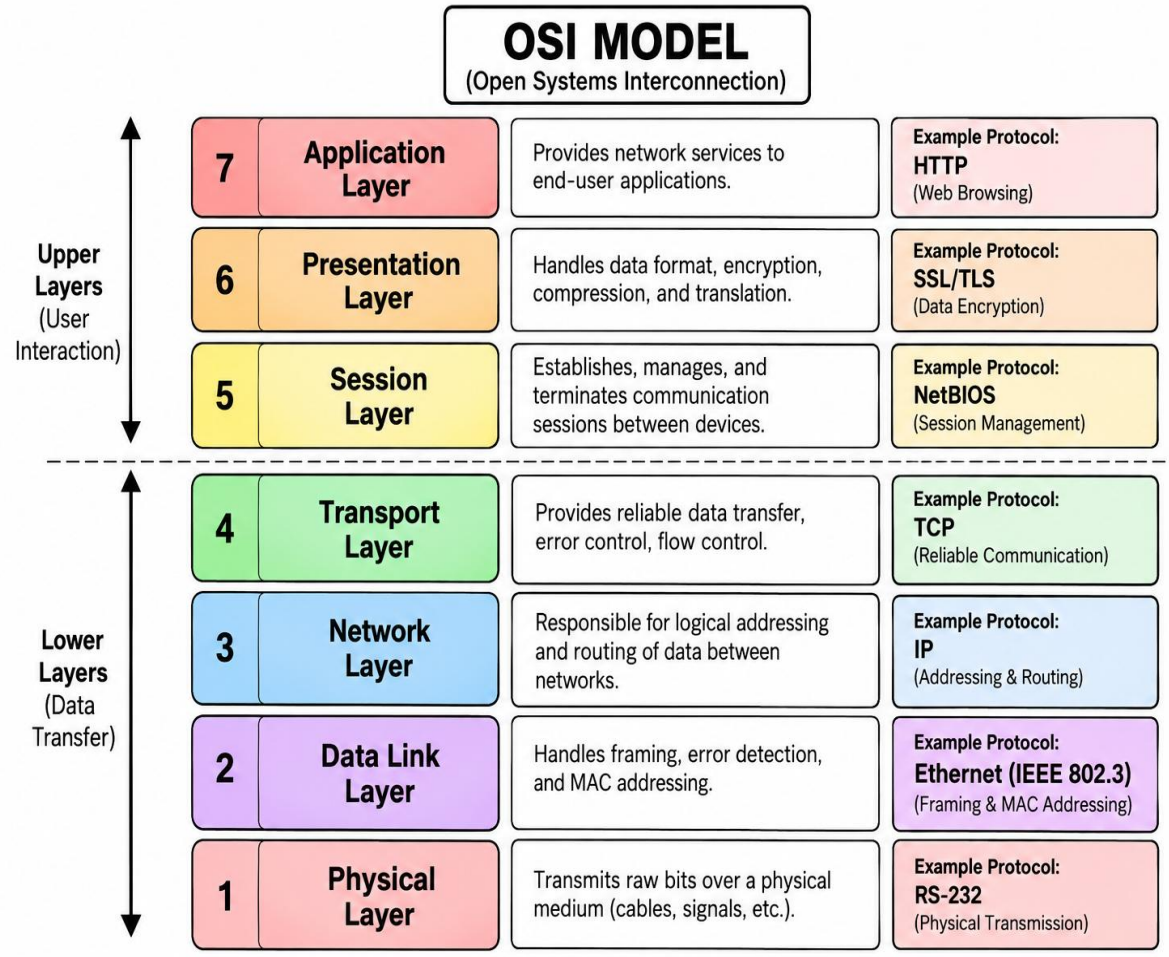
OSI & Network protocols

Tasks :-

1.Draw the OSI model on paper or using any drawing tool, label all 7 layers, and write one real-world example protocol (like HTTP, TCP, IP, etc.) next to each layer.

Ans :-



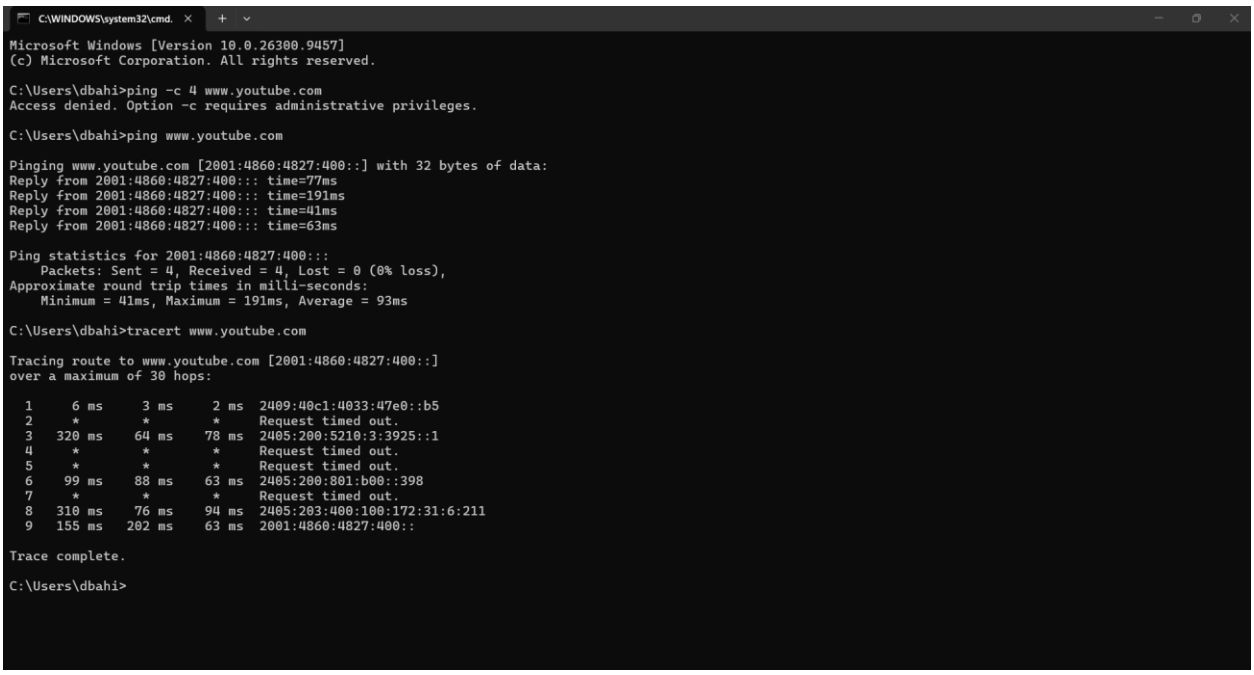
2.Create a table listing 5 popular apps you use (such as WhatsApp, Instagram, Zomato, Flipkart, Paytm) and for each, identify two network protocols that are likely used when you interact with the app (for example, HTTPS, TCP, UDP, DNS, etc.).

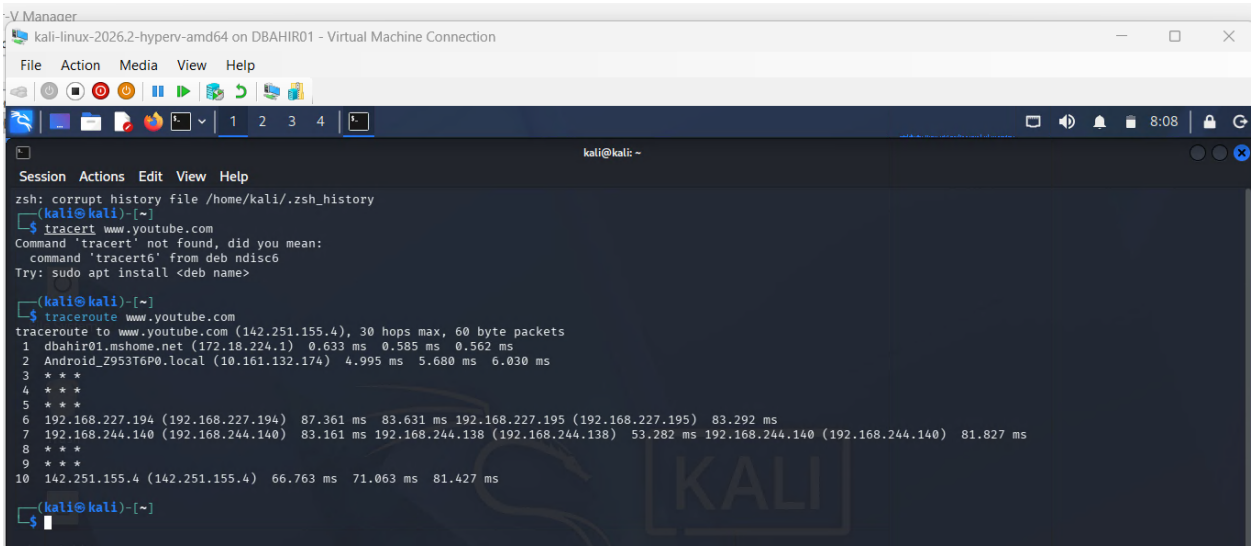
Ans :-

No.	Popular App	Protocol 1	Protocol 2
1	WhatsApp	HTTPS	TCP
2	Instagram	HTTPS	DNS
3	Zomato	HTTPS	TCP
4	Flipkart	HTTPS	DNS
5	Paytm	HTTPS	TLS

3.Use the command prompt or terminal to run 'ping' and 'tracert' (Windows) or 'traceroute' (Linux/Mac) to www.youtube.com, then explain which OSI layers are involved in these commands and how.

Ans :-





- **How Each Command Works Across Layers :-**
    - **ping — Layer 3 focused :**

- **Layer 3:** Your host builds an ICMP Echo Request packet with that IP as destination.

    - **tracert / traceroute — Layer 3 with TTL manipulation :**

- **Layer 3:** The tool sends packets with TTL = 1, then TTL = 2, then TTL = 3, etc.

**4.Write a short scenario describing how a message sent from your device to a friend's device on WhatsApp travels through each OSI layer, mentioning what happens at each step.

Hint: Start from Application layer (your message in WhatsApp) down to Physical layer (actual data sent over WiFi/mobile network), then back up to Application layer on your friend's phone.**

Ans :-

- **Step-by-Step Data Transmission**

OSI Layer	Layer Name	What Happens
7	Application	You type "Hello, how are you?" in WhatsApp and press Send.
6	Presentation	The message data is encoded and protected using encryption, such as TLS for secure transport.
5	Session	Communication between your device and WhatsApp's service is maintained.
4	Transport	TCP or another transport protocol carries the data reliably, depending on the connection.
3	Network	IP addresses are used to route packets across networks toward WhatsApp's servers.
2	Data Link	Data is placed into Wi-Fi or mobile network frames for delivery to the next network device.
1	Physical	Data is transmitted as radio signals over Wi-Fi or the mobile network.

- **Message Reception on Your Friend's Device :-**

OSI Layer	What Happens on Friend's Phone
1. Physical	The phone receives data through Wi-Fi or mobile signals.
2. Data Link	Frames are received and checked for transmission errors.
3. Network	IP packets are processed and delivered to the appropriate device.
4. Transport	Transport data is reassembled and delivered to the appropriate application connection.
5. Session	The communication context is maintained.
6. Presentation	The received data is decrypted and decoded as required.
7. Application	WhatsApp displays the message: "Hello, how are you?"